



mehdi sofiane **DEBBAH**

PHD RESEARCHER IN DEVOPS, AI & 5G

Engineer with a strong background in Artificial Intelligence, IoT, and network systems, with research interests in AI-driven networking, edge computing, privacy, cybersecurity, and the design of next-generation intelligent networks and infrastructures.

✉ mehdi.debbah@hotmail.fr

🏠 Palaiseau (91120)

🚗 Driving licence (B)

📍 France

📞 07 64 46 73 79

SOCIAL NETWORKS

🌐 debbahMehdi

🌐 mehdi-sofiane-debbah

ASSETS

Excellent communication abilities

Strong team spirit

Capacity to work under pressure and manage emergencies

SKILLS

AI & Machine Learning

Diffusion Models, LLMs, GANs, VAEs, TensorFlow & LiteRT, PyTorch, OpenCV.

Cloud & DevOps

GCP, AWS, Docker, Kubernetes, Ansible, CI/CD (GitLab), Terraform, GitHub, Monitoring (Prometheus and Grafana), VMware, LXC.

Networks & Security

Protocols (TCP/IP, OSPF, BGP), VPN, VLAN, Fortinet, IPSec, 5G, Cisco and Amarisoft equipment management.

Programming Languages

C/C++, Python, Java, C#, JavaScript, Bash, PHP.

Web & Mobile Development

HTML5/CSS, Node.js, Angular, Vue.js, WordPress, Flutter, FastAPI.

Database

MySQL, Oracle, Hadoop, Redis.

LANGUAGES

English

C1

French

C2

INTERESTS

Running, basketball, and tennis.

WORK EXPERIENCE

Embedded AI & IoT Research Engineer

From Feb. 2024 to Jun. 2025 INRIA Saclay Île-De-France

- Contribution to the development of an IoT platform for the 5G mMTC project, dedicated to experiments on cellular IoT networks, integrating advanced features such as embedded **AI (TinyML, EdgeAI)** and protocol analysis. **REST API** using Python/FastAPI, Vue.js, and Redis.
- Management LXC VM & containers .
- Configuration of **Cisco** switches for network segmentation using **VLANs**, and implementation of OpenVPN to secure communications between **IoT** devices and the server, ensuring network confidentiality and resilience.
- Deployment of embedded systems for **computer vision** .

R&D Engineer in 5G, Edge Computing, and Augmented Reality

From Mar. 2023 to Sept. 2023 EDF R&D Palaiseau, Île-de-France, France

- Deployment of an Edge Computing infrastructure over a **5G network** for industrial IoT applications, using Amarisoft equipment and a B-Com core network to ensure low latency and high resilience.
- Advanced network configuration with VLANs on Cisco switches, management of sensitive data flows via **VPN and Fortinet firewalls**, ensuring security and service continuity for **IoT** in critical environments.
- Development of an **augmented reality application** on Lenovo A3 smart glasses using Unity and the Qualcomm SDK, optimized for real-time information display in industrial environments.

AI Optimization Python Developer

From July 2020 to Aug. 2020 Fenetch Paris , ile de france , France

- Optimization of the Causal Discovery Toolbox in **Python**, reducing computation time by 30% and improving the accuracy of **causality simulations**.
- Implementation of automated testing to ensure update stability, improving code quality and simplifying maintenance.

WordPress Developer

From July 2018 to Aug. 2018 CentraleSupélec Gif-sur-Yvette, Île-de-France, France

- Design and Management of the RTE Chair website for Université Paris-Saclay in Paris using WordPress.

EDUCATION

PhD in Computer Science, 5G, and Artificial Intelligence

Since Jan. 2026 Télécom Paris, Institut Polytechnique Palaiseau
Energy-Critical Profiling and Compilation for Future Network Infrastructures.
Research on the energy profiling and optimization of network systems and infrastructures, particularly 5G/O-RAN, with a focus on compilation, CPU performance, and energy consumption.

Master in Network and Systems Engineering (IRS)

From Sept. 2021 to Sept. 2023 Université Paris-Saclay Paris, France
Cryptography, IP Protocols, Cloud & Containerization, Network Security.

Master in Distributed Networks and Systems

From Sept. 2019 to Sept. 2021
Houari Boumediene University of Science and Technology (USTHB) Algiers, Algeria
Networks and Protocols, Client/Server Administration, Distributed Applications.

CERTIFICATIONS & ADDITIONAL TRAINING

- [Perform Foundational Infrastructure Tasks in Google Cloud](#)
 - [Set Up and Configure a Cloud Environment in Google Cloud](#)
 - [Create and Manage Cloud Resources](#)
 - **Paris GenAI School – Generative AI Autumn School, 2024**
Intensive program focused on the latest advances in Generative Artificial Intelligence.
 - **Hi! PARIS Summer School HEC – AI & Data Science Summer School, 2024** Training in advanced machine learning, artificial intelligence, and data science.
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